

Text-based transfer function design for semantic volume rendering

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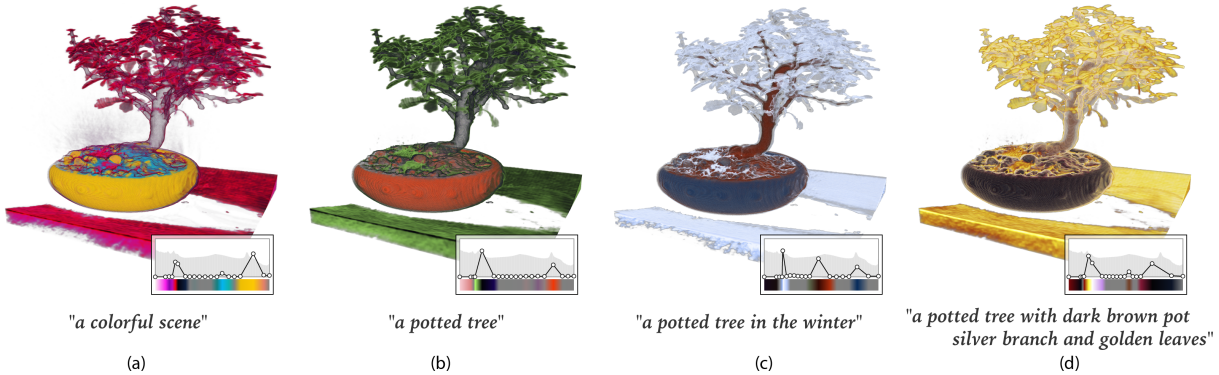


Figure 1: Our method allows users to specify, via natural language, what they wish to see in a volumetric dataset, and in response we optimize for a transfer function that gives volume-rendered images compatible with their description. In this example, we show how (a) a general description permits exploratory analysis of what is contained in a volume, while more precise descriptions lead to TFs that capture the underlying semantics, ranging from (b) natural colors assigned to trees when specifying “potted tree”, (c) snow in the case of “winter”, and (d) detailed specification of visual properties being reflected in the volume-rendered image.

ABSTRACT

Transfer function design is crucial in volume rendering, as it directly influences the visual representation and interpretation of volumetric data. However, creating effective transfer functions that align with users’ visual objectives is often challenging due to the complex parameter space and the semantic gap between transfer function values and features of interest within the volume. In this work, we propose a novel approach that leverages recent advancements in language-vision models to bridge this semantic gap. By employing a fully differentiable rendering pipeline and an image-based loss function guided by language descriptions, our method generates transfer functions that yield volume-rendered images closely matching the user’s intent. We demonstrate the effectiveness of our approach in creating meaningful transfer functions from simple descriptions, empowering users to intuitively express their desired visual outcomes with minimal effort. This advancement streamlines the transfer function design process and makes volume rendering more accessible to a wider range of users.

Index Terms: Transfer function design, vision-language model

1 INTRODUCTION

Volume rendering remains foundational to the visual analysis of volumetric scalar fields. This is in large part due to the interactions offered by volume rendering systems, often taking the form of transfer function (TF) design [21]. Specifically, users control what they see in the volume by specifying how scalar values map to optical properties, namely opacity and color. TF design enables a variety of analyses, ranging from open-ended exploration to help understand what exists within a volume, to more detailed analyses that involve exploring specific objects, materials, or features [18].

However, designing TFs can be a tedious process, particularly for users who have little expertise with volume rendering systems. In response, numerous methods have been proposed for (semi-) automating the generation of TFs [24, 39, 31, 23]. These methods often seek to segment a derived feature space of the volume, for example by using the joint distribution of scalar values and gradient magnitude [15], in order to clearly convey well-delineated objects in the rendered image. Such methods can be effective if the TFs depict structures of interest to the user; otherwise, it becomes necessary for the user to edit the resulting TFs by either refining the opacity or assigning appropriate colors to materials. Numerous methods have been proposed to make TF design more meaningful [36, 32, 34, 5], where users can design TFs with respect to high-level semantics. Yet these methods often require labeled examples provided by users, e.g. data annotated with pre-defined classes [32, 5, 14], or image-based sketching annotations [36, 5], from which to build statistical models of specific objects.

In this paper, we advocate for more intuitive ways to perform semantic TF design, whereby users can *say* what they want to see in the volume, and a TF that is compatible with their description is produced. We introduce text-to-transfer function (T2TF), a method for designing TFs via the specification of natural language, namely a text prompt. T2TF aims to support a variety of analyses common to volume rendering [18]. For instance, when the user is unclear about what might exist in the volume, they can provide generic descriptions to obtain a general understanding of the data (c.f. Fig. 1(a)). From such an overview, a user might wish to be more precise about the objects within a volume, and their visual appearance, and thus they may provide a more detailed description (c.f. Fig. 1(b)-(d)). The specification of natural language allows the design of (1) opacity TFs that are intended to represent user-desired structures, and (2) color TFs that represent the visual appearance of such structures. For instance, when specifying the season of winter for depicting a tree in a given volume, we obtain faded white/brown colors of a tree, reflecting the characteristics of winter (c.f. Fig. 1(c)).

In order to find a TF that satisfies a user’s description, we combine vision-language models, namely CLIP [30], with differentiable volume rendering [40] to optimize for TFs. The most straightforward approach to the problem is to find a TF such that the CLIP-based similarity between the volume-rendered image, and the user’s description, is maximized. However, this is a highly under-constrained optimization problem, as *many* possible TFs, and volume-rendered images, can give high alignment with the description, as measured by CLIP. T2TF addresses these challenges by (1) better constraining optimization, (2) permitting control on the density of rendered materials, and (3) ensuring that the found TFs are interpretable. In detail, inspired by contrastive representation learning [4, 30], our optimization objective ensures that volume-rendered images are well-aligned with the user’s description, while remaining poorly aligned with random descriptions. Secondly, in order to control for the density of materials depicted in volume-rendered images, we incorporate a ray-based prior on transmittance [22]. Last, we introduce a regularization on TFs to minimize their complexity, namely that TFs should only be as complex as necessary. This regularization is motivated by the intended use case of our method, where the output TF should be editable by the user, and thus must be easy to understand.

We demonstrate the benefits of T2TF on a variety of volumes, spanning a variety of text descriptions. We show how descriptions that reference different objects in a volume can lead to TFs that faithfully identify the relevant objects, as largely indicated by opacity (c.f. Fig. 4). Moreover, we show how color TFs tend to reflect the semantics of user descriptions, all achieved without having to explicitly mention colors (c.f. Fig. 1(b)). We further show how our method can support analysis at varying levels of detail, ranging from the specification of general prompts that are intended to give an overview, to more precise prompts intended to depict detailed information within volumes.

In summary, our contributions consist of: (1) an approach to semantics-based TF design based on the specification of natural language, (2) a means of controlling material density in optimized TFs, (3) ensuring TFs are of minimum complexity, thus facilitating post-optimization editing of TFs, and (4) the generality of our method across a large number of volumes and text descriptions.

2 BACKGROUND

The problem we address in this paper can be summarized as follows: given a volumetric scalar field, and a text-based description of the volume that is of interest to the user, the objective is to find a TF whose corresponding volume-rendered images can be accurately described by the user’s text. To solve this problem, T2TF draws from two main research areas: volume rendering, and vision-language models (VLM). To help understand the details of our approach, we first provide the necessary background on both areas.

Volume rendering and TF optimization Our method relies on the absorption-plus-emission model of volume rendering [25]. Specifically, given a scalar field $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ and a viewing ray $\mathbf{v}(t)$ with length parameter t , for which the entry point of the volume is $t = 0$ and exit point $t = T$, the volume rendering equation is:

$$\mathbf{C}(\mathbf{v}; \phi) = \int_0^T T(\mathbf{v}, t; \phi) \sigma_\phi(f_{\mathbf{v}}(t)) \mathbf{c}_\phi(f_{\mathbf{v}}(t)) dt, \quad (1)$$

where $f_{\mathbf{v}}(t) = f(\mathbf{v}(t))$ is the value of the scalar field evaluated at the location $\mathbf{v}(t)$ along the ray, and the functions σ_ϕ and $\mathbf{c}_\phi$ are respectively the density and color TFs, namely 1D functions that depend on parameters ϕ . The function T represents transmittance, or the probability that light along a ray traversing the volume from entry (0) to some current distance (t) will not be absorbed by light:

$$T(\mathbf{v}, t; \phi) = \exp\left(-\int_0^t \sigma_\phi(f_{\mathbf{v}}(u)) du\right) \quad (2)$$

Numerical integration to approximate the integral [25] leads to the following expressions for computing color and transmittance:

$$\mathbf{C}(\mathbf{v}; \phi) \approx \sum_{n=1}^N T_n(\mathbf{v}; \phi) (1 - \exp(-\sigma_\phi(f_{\mathbf{v}}(t_n))\delta)) \mathbf{c}_\phi(f_{\mathbf{v}}(t_n)), \quad (3)$$

$$T_n(\mathbf{v}; \phi) = \prod_{i=1}^n \exp(-\sigma_\phi(f_{\mathbf{v}}(t_i))\delta), \quad (4)$$

where N represents the number of steps taken along the ray, t_n the ray distance for step n , and δ the step size

Notably, if density and color TF access are both differentiable functions, then the color in Eq. 3 can be viewed as a differentiable function in the TF (namely, its parameters ϕ), and camera pose. This has led to numerous methods for finding rendering parameters via gradient-based optimization of volume rendering. For instance, Weiss et al. [40] optimize TFs to match an image reference, further extended towards the comparison of fields [26], as well as the exploration of a collection of TFs [28]. For T2TF, however, we do not have a reference image on which to compute an image-based loss. Rather, given a text description, we require a means of scoring how well a volume-rendered image matches the user’s text.

Vision-language models Recently, a number of foundation models have been developed to align images with text [30, 3, 38]. Specifically, the model of CLIP [30] has shown that when trained on a massive amount of data, namely hundreds of millions to billions [33] of captioned images, high-capacity neural networks are capable of accurately scoring whether a given image is well-described by a text prompt. This is beneficial for T2TF in the following way: given a volume-rendered image denoted $\mathbf{I}$, and a user-provided text description y , we can compute a vector-based encoding of both the image $\mathbf{e}_I(\mathbf{I})$ and text $\mathbf{e}_y(y)$, and their dot product $\mathbf{e}_I(\mathbf{I})^T \mathbf{e}_y(y)$ gives us a score of image-text alignment. At a basic level, T2TF seeks a TF such that its corresponding volume-rendered image yields a high CLIP score with the user’s text.

A major motivation for us in using CLIP is its ability to perform zero-shot categorization of images [27], demonstrating a deep understanding of both visual features and natural language [8]. However, despite its robustness, CLIP nevertheless gives reduced performance under distribution shift [35]. Specific to our problem setting, we anticipate covariate shift due to the simplified illumination model (c.f. Eq. 3), for instance, the rendered tree in Fig. 1 appears different from an actual tree. Handling such shortcomings has been addressed in the computer vision community, namely for the purposes of generating 3D radiance fields [11, 29, 19, 37] solely from a given text description. In particular, our work is motivated by prior work [11, 29, 37], in which a radiance field, e.g. a neural network that reports density and color at each location, is optimized such that its volume-rendered images either give high CLIP scores [11], or high likelihood with respect to a diffusion model [29, 37].

3 TEXT-BASED TRANSFER FUNCTION DESIGN

In this section we describe our method for mapping text descriptions to TFs, please see Fig. 2 for a graphical overview, and Algorithm 1 for a description of our optimization procedure. We defer additional details on optimization to supplementary material.

TF representation We choose to represent both density (σ_ϕ) and color ($\mathbf{c}_\phi$) TFs as piecewise linear functions. Thus, TF parameters ϕ are comprised of (1) control points, restricted to the range of f , and for each control point (2) a nonnegative value representing density, and (3) a 3-vector representing the color in RGB space, each entry constrained to $[0, 1]$. The number of control points is not a concern for computational efficiency or differentiation, as the piecewise-linear function is sampled to a fine grid initially and directly accessed during volume rendering, ensuring the loss remains differentiable in the control point locations and values. We chose

Algorithm 1: Text-to-transfer function

Input : Scalar field f , text description y
Output: Parameters ϕ for density TF σ_ϕ and color TF $\mathbf{c}_\phi$
 $\mathbf{c}_\phi$: random initialization, $\sigma_\phi : \text{ave}(T_N(\mathbf{V}_0; \phi)) \approx \rho$;
for $i \leftarrow 1$ **to** N **do**
 $\mathbf{V} \sim \mathcal{V}$ (sample camera pose);
 $\mathbf{B} \sim \mathcal{B}$ (sample background image);
 $\tilde{\mathbf{y}} \sim \mathcal{P}$ (sample negative prompts);
 $\mathbf{I} \leftarrow \mathbf{C}(\mathbf{V}; \phi) + T_N(\mathbf{V}; \phi)\mathbf{B}$;
 $\mathcal{L}_{\text{CLIP}}(\phi) \leftarrow CE(\mathbf{e}_I(\mathbf{I})^T \mathbf{e}_y(y); \mathbf{e}_I(\mathbf{I})^T \mathbf{e}_y(\tilde{\mathbf{y}}))$;
 $\mathcal{L}_{\text{density}}(\phi) \leftarrow -\log P_{\text{beta}}(T_N(\mathbf{V}; \phi); a, b)$;
 $\mathcal{L}_{\text{reg}}(\phi) \leftarrow \lambda_1 \|\sigma_\phi\|_1 + \lambda_2 \|\mathbf{c}_\phi - 0.5\|_2^2$;
 $\phi \leftarrow \phi - \eta \nabla_\phi (\mathcal{L}_{\text{CLIP}}(\phi) + \mathcal{L}_{\text{density}}(\phi) + \mathcal{L}_{\text{reg}}(\phi))$;
end

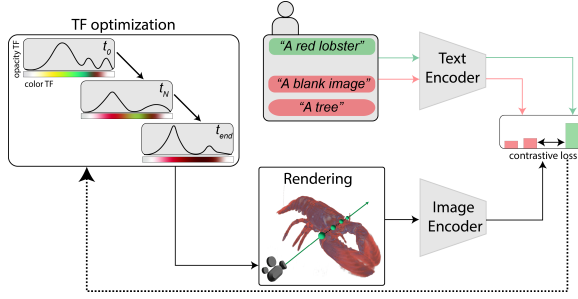


Figure 2: We show an overview of T2TF. Given a TF, we volume render from a camera view to produce an image and subsequently obtain a CLIP-based encoding of the image. We then compare the image with a CLIP-based encoding of the user’s prompt, expressive of their visualization objective, with negative sampled prompts. We form a contrastive loss between the user’s prompt and negative prompts, giving us a way to update the TF via gradient-based optimization.

this representation due to its predominance in scientific visualization software [1, 6], and its ease of editing TFs via dragging control points, post-optimization.

Objective Our goal is to find a TF that satisfies the user’s description in the volume-rendered output, while also allowing for control on the density of rendered materials, and last, ensuring the found TF is of low complexity. Our objective thus follows as a combination of three loss terms addressing each of these considerations, wherein we aim to minimize:

$$\mathcal{L}(\phi) := \mathcal{L}_{\text{CLIP}}(\mathbf{I}(\mathbf{V}, \mathbf{B}; \phi)) + \mathcal{L}_{\text{density}}(T_N(\mathbf{V}; \phi)) + \mathcal{L}_{\text{reg}}(\phi). \quad (5)$$

Specifically, for a given camera pose $\mathbf{V}$ and background image $\mathbf{B}$, we denote $\mathbf{I}(\mathbf{V}, \mathbf{B}; \phi)$ as the volume-rendered image. The term $\mathcal{L}_{\text{CLIP}}$ represents a CLIP-based loss, intended to align the image with the user’s description. Denoting $T_N(\mathbf{V}; \phi)$ as the transmittance for each viewing ray associated with pose $\mathbf{V}$, the term $\mathcal{L}_{\text{density}}$ controls for material density via placing a prior on transmittance. Last, the term $\mathcal{L}_{\text{reg}}(\phi)$ is a complexity penalty on the TF. We optimize over a distribution of camera poses $\mathbf{V} \sim \mathcal{V}$, and background images $\mathbf{B} \sim \mathcal{B}$, in order to ensure that the user’s description is satisfied under multiple views, while also ensuring that CLIP does not overfit to the background, following prior work [11, 29]. Prior to optimization, the color TF is randomly initialized, while the density TF is found such that average transmittance, at a randomly sampled fixed view $\mathbf{V}_0$, approximately satisfies a specified value ρ . In practice $\rho = .05$, to ensure the field’s full range of values contributes to volume rendering.

CLIP-based learning of TFs A straightforward way to find TFs is to simply maximize the CLIP score of a volume-rendered

image with the user’s description. However, we found this to be often unreliable, emphasizing transparent materials and volume background; please see supplemental for comparisons. As it is possible for the given image to satisfy *many* different descriptions, we aim to find a TF whose volume-rendered images are *unique* to the user’s description. To this end, we propose a contrastive loss [4, 30] for learning TFs. Specifically, at each step of optimization, we sample, uniformly at random, a collection of *negative* prompts from a pre-defined set of image-captions $\mathcal{P}$ – in practice, we use LAION [33]. We then compute the CLIP score between the volume-rendered image, and (1) the user’s prompt y , as well as (2) the collection of sampled negative prompts denoted $\tilde{\mathbf{y}} \sim \mathcal{P}$. We treat these scores as logits, subsequently passed to a softmax to form a discrete distribution, and then compute a cross-entropy loss to discriminate a user’s description from negative prompts. We find little difference between experiments run with different number of negative prompts.

We can further extend our formulation to help the user *suppress* certain structures in the output; this is most useful when it is challenging to include these omissions in the given description. Specifically, when we sample negative prompts $\tilde{\mathbf{y}}$, we append the user’s supplied set of negative prompts. As shown in Fig. 4(d), user-supplied negative prompting can help omit undesirable materials, for instance skin, skull, to capture just one material, namely teeth.

Density prior In controlling for material properties, not everything can be so easily spoken; this is especially the case for material density, which is continuous, and thus difficult to express in language. To this end, following Lombardi et al. [22], we place a beta prior on transmittance $T_N(\mathbf{V}; \phi)$, with prior parameters $a = b < 1$ in order to give a bias towards the transmittance being either 0 or 1. This is helpful in biasing discovered materials of the TF to have higher density, while also discouraging semi-transparency, as viewing rays whose transmittance is small are biased towards zero transmittance. We found this prior to give consistent results, in comparison to other methods considered for neural radiance fields [11, 13]; please see supplemental for comparisons.

TF regularization Although the TF that we find will have high density placed in relevant portions of the volume’s range space, unless appropriately regularized, it might also place high density in portions that have little impact on the volume-rendered results, or occluded by the relevant material. If one wants to edit the resulting TF, this can lead to some confusion about what is relevant to edit. To this end, we aim to minimize the complexity of the TF, wherein we apply a sparsity-promoting l_1 regularization to the density evaluated at the control points. This will encourage low-density assignment to function values not relevant to the materials displayed. Moreover, for the color TF, we additionally apply an l_2 regularization towards gray colors; this is done primarily for optimization stability purposes, so that colors do not saturate at the extremes (e.g. 0 or 1) over the course of optimization.

4 RESULTS

This section demonstrates the images produced from T2TF transfer functions on multiple datasets. The datasets are from Open Scientific Visualization Datasets through OpenVisus. Details of each dataset are provided in the supplementary material.

We have demonstrated in Fig. 1 that T2TF can produce different color TFs that match the theme of the prompt. In Fig. 4, T2TF is asked to create images highlighting various features of the human head from the visible male dataset, a CT scan of a human head. Fig. 4(a) results from a generic prompt asking for a CT scan of a human head. The found TF gives an overview of the scene, depicting two large features: a transparent skin and a solid skull, shown as two peaks in the opacity TF. For a more precise description, in Fig. 4(b) we request the skin of the scan. We use the word “clean” to reduce ambient noise, which has similar scalar values to the skin, and we find the single peak opacity TF effectively renders the skin.

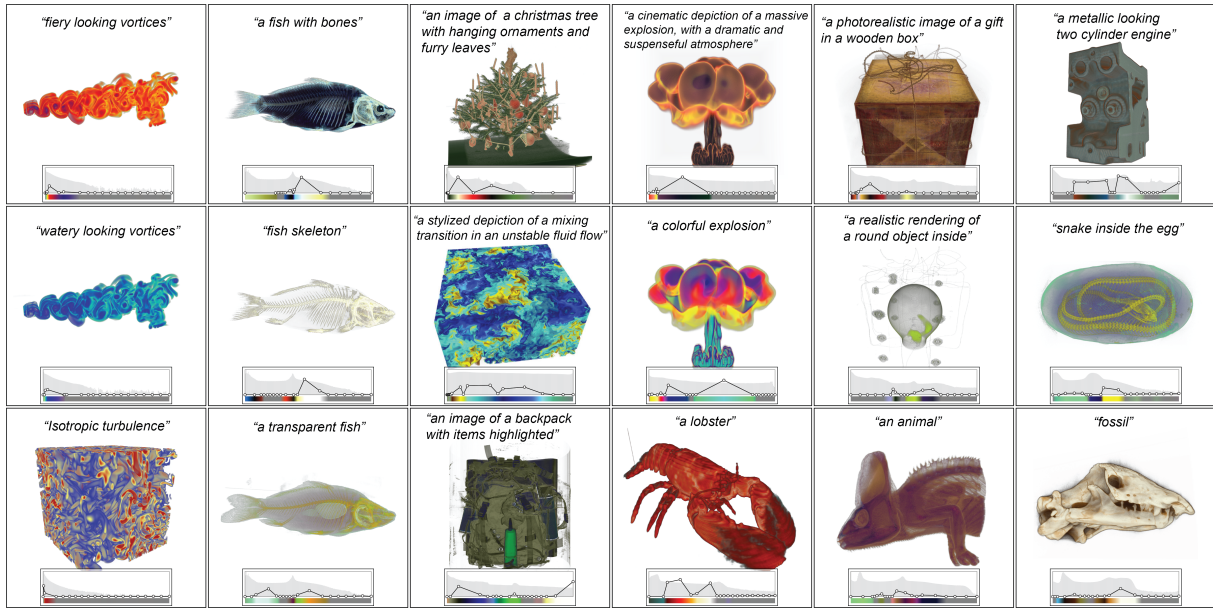


Figure 3: T2TF results for a variety of volumes (scanned data, numerical simulations) and prompts (general vs. detailed descriptions).

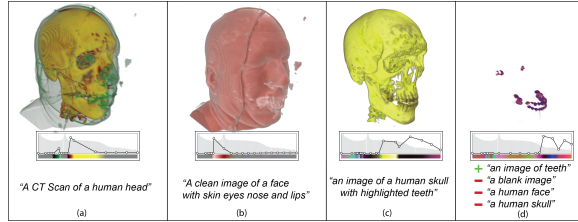


Figure 4: Volume-rendered images of the visible male dataset. In (a), the generic prompt results in the transfer functions that show an overview of features in the volume. Then, in (b)-(d), different opacity transfer functions highlight the skin and skull. This example illustrates the capability of T2TF to adjust opacity transfer functions

In Fig. 4(c) request an image of both the skull and the teeth, where we find the resulting TF contains two peaks in the opacity TF. These peaks correspond to the scalar values associated with the skull and teeth, while other regions are mostly flat. The color TF uses a contrasting color to the skull, in order to highlight the teeth. Fig 4(d) shows our setup to highlight one small feature, the teeth, of the CT scan. We found it is most effective to add negative prompts to remove the large feature when we want to isolate smaller features in the scene. To this end, we provide three additional negative prompts: “a human skull” and “a human face”, in order to suppress these features, and “a blank image”, to ensure the TF does not degenerate to a trivial solution. As shown, the opacity TF only peaks at the scalar value representing the teeth in the dataset.

In Figure 3, we showcase our approach by providing an assortment of T2TF-generated transfer functions and their volume-rendered images. The results suggest that T2TF can produce meaningful TFs for a diverse set of volumetric data, ranging from scanned data to numerical simulations, and different types of prompts, from general to more detailed descriptions of scenes.

5 DISCUSSION

In this work, we have introduced T2TF, a method for designing TFs via the specification of natural language. Although our results show promise for this new way of semantically specifying TFs, we acknowledge several limitations of our approach. First, the computational cost required to optimize for a TF places our method at

far from interactive. The complexity is largely dominated by the cost to backpropagate through the volume renderer, and this further depends on the size of the volume. Specifically, for the visible male volume, it takes approximately 0.5s for a single step of gradient descent for a single camera pose, while for the Rayleigh-Taylor instability volume, this takes around 2s per step. For our results, we conservatively set the number of optimization steps in order to ensure convergence, namely 300 steps, but in practice, we find that the number of steps can be reduced while still obtaining similar results; future work will investigate trade-offs between speed and quality. Secondly, we assume that what the user specifies in their description is visible at arbitrary camera poses, as well as the initial camera pose for computing average transmittance. However, for features that are more localized, and thus view-specific, this assumption may no longer hold. A method for viewpoint optimization, similar to Weiss et al. [40], is therefore required to address this issue.

Our work opens doors to many future directions. In many ways, our work is complementary to the extensive body of research in volume rendering, and thus we intend to explore how multidimensional TFs [16, 17], illumination models [10], and image-based TF editing [9] can enhance T2TF. Moreover, we need not limit ourselves to CLIP, and can instead use proper likelihood-based models, namely text-conditional diffusion models [29, 37]. More broadly, we will investigate multi-modal large language models [20] for T2TF, as they demonstrate stronger language understanding and better vision-language alignment. Lastly, we used prompts known to work better based on our experience. However, a detailed prompt may result in a poor local minimum. To improve on optimization, we plan on exploring methods that iteratively refine TFs through a coarse-to-fine prompting scheme, whereby a TF found for a general description is used as initialization in finding a TF for a more precise description.

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REFERENCES

- [1] U. Ayachit. *The paraview guide: a parallel visualization application*. Kitware, Inc., 2015. 3
- [2] I. Baeza Rojo and T. Günther. Vector field topology of time-dependent flows in a steady reference frame. *IEEE Trans. Vis. Comput. Graph. (Proc. IEEE Scientific Visualization)*, 2019. 1
- [3] H. Bao, W. Wang, L. Dong, Q. Liu, O. K. Mohammed, K. Aggarwal, S. Som, S. Piao, and F. Wei. Vlm0: Unified vision-language pre-training with mixture-of-modality-experts. *Advances in Neural Information Processing Systems*, 35:32897–32912, 2022. 2
- [4] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton. A simple framework for contrastive learning of visual representations, 2020. 2, 3
- [5] H.-C. Cheng, A. Cardone, S. Jain, E. Krokos, K. Narayan, S. Subramaniam, and A. Varshney. Deep-learning-assisted volume visualization. *IEEE Trans. Vis. Comput. Graph.*, 25(2):1378–1391, 2018. 1
- [6] H. Childs, E. Brugger, B. Whitlock, J. Meredith, S. Ahern, D. Pugmire, K. Biagas, M. Miller, C. Harrison, G. H. Weber, et al. Visit: An end-user tool for visualizing and analyzing very large data. 2012. 3
- [7] A. W. Cook, W. Cabot, and P. L. Miller. The mixing transition in Rayleigh-Taylor instability. *Journal of Fluid Mechanics*, 511:333–362, 2004. doi: 10.1017/S0022112004009681 2
- [8] W. Dai, L. Hou, L. Shang, X. Jiang, Q. Liu, and P. Fung. Enabling multimodal generation on clip via vision-language knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2022*, pp. 2383–2395, 2022. 2
- [9] H. Guo, N. Mao, and X. Yuan. Wysiwyg (what you see is what you get) volume visualization. *IEEE Trans. Vis. Comput. Graph.*, 17(12):2106–2114, 2011. 4
- [10] F. Hernell, P. Ljung, and A. Ynnerman. Local ambient occlusion in direct volume rendering. *IEEE Trans. Vis. Comput. Graph.*, 16(4):548–559, 2009. 4
- [11] A. Jain, B. Mildenhall, J. T. Barron, P. Abbeel, and B. Poole. Zero-shot text-guided object generation with dream fields. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 867–876, 2022. 2, 3, 1
- [12] A. Kanitsar, T. Theußl, L. Mroz, M. Šrámek, A. V. Bartrolí, B. Cséfalvi, J. Hladůvka, D. Fleischmann, M. Knapp, R. Wegenkittl, P. Felkel, S. Röttger, S. Guthe, W. Purgathofer, and M. E. Gröller. Christmas tree case study: Computed tomography as a tool for mastering complex real world objects with applications in computer graphics. In *IEEE Visualization, 2002. VIS 2002.*, pp. 489–492, 2002. doi: 10.1109/VISUAL.2002.1183812 2
- [13] M. Kim, S. Seo, and B. Han. Infonerf: Ray entropy minimization for few-shot neural volume rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12912–12921, 2022. 3, 1
- [14] S. Kim, Y. Jang, and S.-E. Kim. Image-based tf colorization with cnn for direct volume rendering. *IEEE Access*, 9:124281–124294, 2021. 1
- [15] G. Kindlmann. Semi-automatic generation of transfer functions for direct volume rendering. Master’s thesis, Cornell University, USA, 1999. 1
- [16] J. Kniss, G. Kindlmann, and C. Hansen. Interactive volume rendering using multi-dimensional transfer functions and direct manipulation widgets. In *Proceedings Visualization, 2001. VIS’01.*, pp. 255–262. IEEE, 2001. 4
- [17] J. Kniss, G. Kindlmann, and C. D. Hansen. Multidimensional transfer functions for volume rendering. In C. D. Hansen and C. R. Johnson, eds., *Visualization Handbook*, pp. 189–209. Butterworth-Heinemann, 2005. 4
- [18] B. Laha, D. A. Bowman, D. H. Laidlaw, and J. J. Socha. A classification of user tasks in visual analysis of volume data. In *2015 IEEE Scientific Visualization Conference (SciVis)*, pp. 1–8. IEEE, 2015. 1
- [19] H.-H. Lee and A. X. Chang. Understanding pure clip guidance for voxel grid nerf models. *arXiv preprint arXiv:2209.15172*, 2022. 2
- [20] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36, 2024. 4
- [21] P. Ljung, J. Krüger, E. Groller, M. Hadwiger, C. D. Hansen, and A. Ynnerman. State of the art in transfer functions for direct volume rendering. In *Computer graphics forum*, vol. 35, pp. 669–691. Wiley Online Library, 2016. 1
- [22] S. Lombardi, T. Simon, J. Saragih, G. Schwartz, A. Lehrmann, and Y. Sheikh. Neural volumes: Learning dynamic renderable volumes from images. *ACM Trans. Graph.*, 38(4):65:1–65:14, July 2019. 2, 3, 1
- [23] B. Ma and A. Entezari. Volumetric feature-based classification and visibility analysis for transfer function design. *IEEE Trans. Vis. Comput. Graph.*, 24(12):3253–3267, 2017. 1
- [24] R. Maciejewski, I. Woo, W. Chen, and D. Ebert. Structuring feature space: A non-parametric method for volumetric transfer function generation. *IEEE Trans. Vis. Comput. Graph.*, 15(6):1473–1480, 2009. 1
- [25] N. Max. Optical models for direct volume rendering. *IEEE Trans. Vis. Comput. Graph.*, 1(2):99–108, June 1995. doi: 10.1109/2945.468400 2
- [26] C. Neuhauser and R. Westermann. Transfer function optimization for comparative volume rendering. *arXiv preprint arXiv:2306.05885*, 2023. 2
- [27] Z. Novack, J. McAuley, Z. C. Lipton, and S. Garg. Chils: Zero-shot image classification with hierarchical label sets. In *International Conference on Machine Learning*, pp. 26342–26362. PMLR, 2023. 2
- [28] B. Pan, J. Lu, H. Li, W. Chen, Y. Wang, M. Zhu, C. Yu, and W. Chen. Differentiable design galleries: A differentiable approach to explore the design space of transfer functions. *IEEE Trans. Vis. Comput. Graph.*, 2023. 2
- [29] B. Poole, A. Jain, J. T. Barron, and B. Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv preprint arXiv:2209.14988*, 2022. 2, 3, 4
- [30] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021. 2, 3
- [31] M. Ruiz, A. Bardera, I. Boada, I. Viola, M. Feixas, and M. Sbert. Automatic transfer functions based on informational divergence. *IEEE Trans. Vis. Comput. Graph.*, 17(12):1932–1941, 2011. 1
- [32] C. R. Salama, M. Keller, and P. Kohlmann. High-level user interfaces for transfer function design with semantics. *IEEE Trans. Vis. Comput. Graph.*, 12(5):1021–1028, 2006. 1
- [33] C. Schuhmann, R. Beaumont, R. Vencu, C. Gordon, R. Wightman, M. Cherti, T. Coombes, A. Katta, C. Mullis, M. Wortsman, et al. Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in Neural Information Processing Systems*, 35:25278–25294, 2022. 2, 3, 1
- [34] K. P. Soundararajan and T. Schultz. Learning probabilistic transfer functions: A comparative study of classifiers. In *Computer Graphics Forum*, vol. 34, pp. 111–120. Wiley Online Library, 2015. 1
- [35] W. Tu, W. Deng, and T. Gedeon. A closer look at the robustness of contrastive language-image pre-training (clip). *Advances in Neural Information Processing Systems*, 36, 2024. 2
- [36] F.-Y. Tzeng, E. B. Lum, and K.-L. Ma. An intelligent system approach to higher-dimensional classification of volume data. *IEEE Trans. Vis. Comput. Graph.*, 11(3):273–284, 2005. 1
- [37] H. Wang, X. Du, J. Li, R. A. Yeh, and G. Shakhnarovich. Score jacobian chaining: Lifting pretrained 2d diffusion models for 3d generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12619–12629, 2023. 2, 4
- [38] W. Wang, Z. Chen, X. Chen, J. Wu, X. Zhu, G. Zeng, P. Luo, T. Lu, J. Zhou, Y. Qiao, et al. Visionllm: Large language model is also an open-ended decoder for vision-centric tasks. *Advances in Neural Information Processing Systems*, 36, 2024. 2
- [39] Y. Wang, W. Chen, J. Zhang, T. Dong, G. Shan, and X. Chi. Efficient volume exploration using the gaussian mixture model. *IEEE Trans. Vis. Comput. Graph.*, 17(11):1560–1573, 2011. 1
- [40] S. Weiss and R. Westermann. Differentiable direct volume rendering. *IEEE Trans. Vis. Comput. Graph.*, 28(1):562–572, 2021. 2, 4